

TABLE 1
PROPERTIES OF RECENTLY DEVELOPED
BIFUNCTIONAL AIR ELECTRODES

Catalyst Type	Supporting Material	E_{ORR} V (vs. Hg/HgO)	E_{OER} V	j_{max} ORR at -0.2VmA cm ² (current density)	Lifetime Number of Cycles
Ag	Sintered Ni powder	—	—	-80	500
Ag-MnO ₂ nanorods	Graphite powder	0.0.03	0.61	-0.25	270
Perovskites	Ketjen black glassy carbon	-0.08	0.29	-200	100
Spinel LaMnO ₃	Carbon black	-0.09	0.60	-10	150

Note: Recently, it has been established that adding 1.0mM of an ionic liquid named EML to the electrolyte vastly improves performance. EML suppresses hydrogen evolution and increases the average capacity retention to 94% even after 1000 cycles of charge and discharge.

hydroxides for the OE reaction.

In both spinel and LDH, the surface active compounds are metal hydroxides and oxyhydroxides, which have high conductivity and activity and are stable.

Nanostructured carbon-supported transition metal atoms or metal alloy nanoparticles show good ORR and OER electrocatalysis. Heteroatoms such as N, B, and P can be incorporated into carbon frameworks for enhanced electrocatalytic activity through electron modulation. Moreover, these dopants also serve as anchoring centres to attach metal atoms or metal alloy particles through strong coupling effects, forming electrocatalytically active and stable M-N-C molecules, as shown in Fig. 11.

Cobalt (Co) nanoparticles and N-doped carbon provide electrocatalytic ORR activity, and the network constructed by carbon fibres and nanotubes ensures high electrical conductivity and a porous structure. The development of bifunctional ORR/OER electrocatalysts can be very useful for metal-air batteries' discharge and charge processes. Hybrid material of cobalt-coordinated porphyrin (superb hydrophilic surface with accessibility to electrolyte) covalent organic framework-coated CNTs can be used as a bifunctional catalyst.

Metal oxides like Mn, Co, Ni,

and Fe oxides, can also be used as bifunctional catalysts because of their intrinsic activities, low cost, and structural stability. Integration of metal oxides with nanostructured carbon supports provides an excellent conductive network and accessible surface area to facilitate electron transfer and mass diffusion during ORR and OER.

It has been found that there is another class of efficient and stable bifunctional non-precious metal composite catalysts comprising metal-nitrogen-carbon (MNC) for ORR, and metal oxyhydroxide as OER catalysts. Transition metals containing nitrogen-doped carbon catalysts, especially of the Fe-NC type, have been reported to have excellent ORR activity in both acidic and alkaline electrolytes.

The addition of cobalt leads to a very good performing catalyst with an OER/ORR gap of 0.79V, high output, and high cycle life. The bifunctional behaviour is attributed to an interaction of CoN₄ centres for the ORR and Co-oxide particles transforming to hydroxides for the OER.

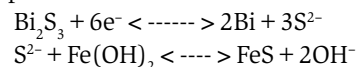
Table 1 shows some of the properties of recently developed bifunctional air electrodes.

Negative iron electrodes

The iron (Fe) electrodes are generally made from powdered materials

that contain metallic iron or iron oxides, most commonly Fe₃O₄, which is supported on conductive material like carbon in order to increase the conductivity and surface area. Instead of using commercial Fe₃O₄ powder for making Fe electrodes, the use of nanosized particles of iron is the best strategy to make more of the solid electrode accessible to the electrolyte, as this step increases the surface area, resulting in more charge capacities. Carbonyl iron electrodes have been found to increase the discharge-specific capacity even though their particle size is not nanosized. The discharge products like iron hydroxides of iron electrodes are insulating in nature and can passivate the electrodes, resulting in further oxidation.

Research has indicated that special carbons like graphene, multi-wall carbon nanotube, and amorphous carbon composites admixed with copper and tin have the potential to enhance iron electrode performance by suppressing hydrogen evolution. A variety of compounds like Bi₂S₃, Bi₂O₃, FeS, and Na₂S are added as pore formers to the Fe electrode to reduce the rate of hydrogen evolution and increase the charging efficiency. In the electrode, pore formers create enough electrolyte access to the reaction site. The best result is obtained when 5% (by wt) Bi₂S₃ is used as an additive. Bismuth sulphide decomposes in a redox reaction by forming iron sulphide as shown below:



Iron sulphide enables a high discharge rate via depassivating the oxide, modifying the electrode kinetics, and increasing the solubility of iron hydroxide, making the reaction fast. The metallic bismuth increases hydrogen overpotential and charging efficiency from 50% to 90%. The presence of sulphide derivatives